Classification: Performance Metrics & Class Imbalance

Big Data y Machine Learning para Economía Aplicada

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Agenda

1. Recap
2. Other Algorithms for Probability Estimation
   - Non-Parametrics: K-Nearest Neighbors
   - Generative Models for Classification
     - Discriminant Analysis
   - Naive Bayes
3. Performance Metrics
   - Accuracy
   - TNR
   - TNR
4. ROC curve
5. Imbalanced Classification
   - Metrics

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Classification: Motivation

- Many predictive questions are about classification
  - Credit, Poverty, Firm default, Fraud, Unemployment, etc.
- Aim is to classify y, where y represents membership in a category
  - Qualitative, not necessarily ordered
  - We will focus for now in the binary case

The prediction question is, given a new X,
what is our best guess at the response category $\hat{y}$

Classification: Recap

$$\hat{y}_i = 1[p_i \geq c]$$

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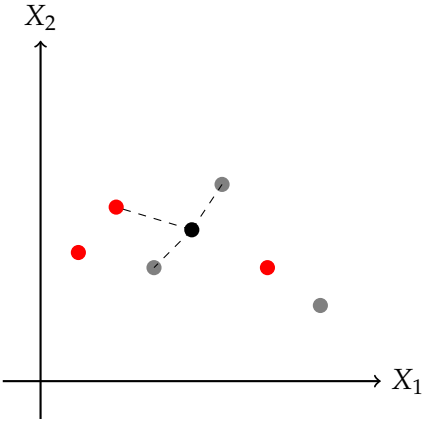
K-Nearest Neighbors

- K nearest neighbor (K-NN) algorithm predicts class $\hat{y}$ for x by asking *What is the most common class for observations around x?*



K-Nearest Neighbors

- K nearest neighbor (K-NN) algorithm predicts class $\hat{y}$ for x by asking *What is the most common class for observations around x?*
- Algorithm: given an input vector x_f where you would like to predict the class label
  - Find the K nearest neighbors in the dataset of labeled observations, $\{x_i, y_i\}_{i=1}^{n}$, the most common distance is the Euclidean distance:

$$d(x_i, x_f) = \sqrt{\sum_{j=1}^{p} (x_{ij} - x_{fj})^2} \tag{1}$$

  - This yields a set of the K nearest observations with labels:

$$[x_{i1}, y_{i1}], \ldots, [x_{iK}, y_{iK}] \tag{2}$$

  - The predicted class of x_f is the most common class in this set

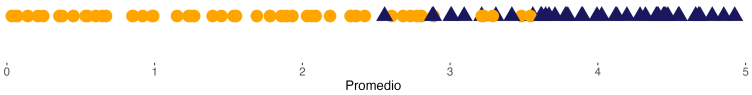
$$\hat{y}_f = mode\{y_{i1}, \ldots, y_{iK}\} \tag{3}$$

Generative Models for Classification

- LDA
- QDA
- Naive Bayes

Discriminant Analysis

$$Pr(Y = 1|X) \tag{4}$$



Linear Discriminant Analysis

Reverend Bayes to the rescue: Bayes Theorem

Discriminant Analysis

Linear Discriminant Analysis

Example: Unemployment

photo from https://www.dailydot.com/parsec/batman-1966-labels-tumblr-twitter-vine/

Linear Discriminant Analysis

Extensions

- If we have k predictors?
- then $X|Y \sim NM(\mu, \Sigma)$

$$f(X|Y=j) = \frac{1}{(2\pi)^{k/2}|\Sigma|^{1/2}} exp(-\frac{1}{2}(x-\mu_j)'\Sigma_j(x-\mu_j) \tag{5}$$

- μ_j is the vector of the sample means in each partition $j = 0, 1$
- Σ_j is the matrix of variance and covariances of each partition $j = 0, 1$

Linear Discriminant Analysis

- Why is it call linear?
- Note

$$p_i > \frac{1}{2} \iff ln(\frac{p_i}{(1-p_i)}) \tag{6}$$

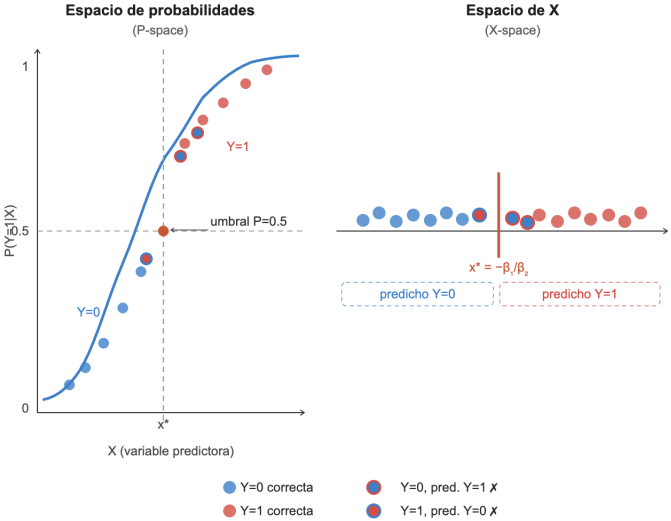
- Logit with one predictor

$$\beta_1 + \beta_2 X \tag{7}$$

- Classification: in the probability of space
- Discrimination: in the space of X
- $\beta_1 + \beta_2 X$ is the discrimination function for logit (it is lineal)

Linear Discriminant Analysis

Discrimination function



Linear Discriminant Analysis

- LDA?
- One predictor with $\sigma_0 = \sigma_1$ (equal variance)

$$Pr(Y=1|X) = \frac{f(X|Y=1)Pr(Y=1)}{f(X|Y=1)Pr(Y=1) + f(X|Y=0)(1-Pr(Y=1))} \tag{8}$$

- Then under the equal variance assumption

$$\frac{Pr(Y=1|X)}{1-Pr(Y=1|X} = \frac{f(X|Y=1)Pr(Y=1)}{f(X|Y=0)(1-Pr(Y=1))} \tag{9}$$

$$= \frac{Pr(Y=1)exp((x-\mu_1)^2)}{(1-Pr(Y=1))exp((x-\mu_0)^2)} \tag{10}$$

Linear Discriminant Analysis

- Taking logs

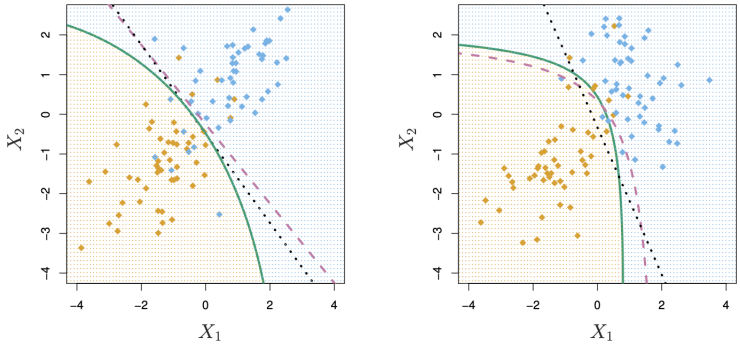
$$
\begin{aligned}
log\left(\frac{Pr(Y=1|X)}{1-Pr(Y=1|X}\right) &= log(\frac{Pr(Y=1)}{(1-Pr(Y=1))} + (x-\mu_1)^2 - (x-\mu_0)^2 && (11)\\
&= log(\frac{Pr(Y=1)}{(1-Pr(Y=1))} + \mu_1^2 - \mu_0^2 - 2(\mu_1-\mu_0)x && (12)\\
&= \gamma_1 + \gamma_2 X && (13)
\end{aligned}
$$

- under the assumption of equal variance the discrimination function is lineal

Discriminant Analysis

Quadratic Discriminant Analysis

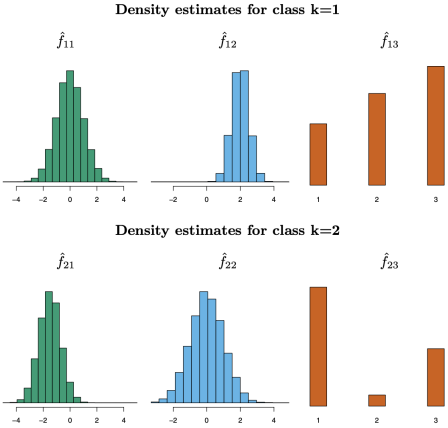
- QDA assumes diferent variances for the components



Naive Bayes

- NB assumes independence

$$f(X|Y=1) = f(x_1|Y=1) \times \cdots \times f(x_k|Y=1) \tag{14}$$



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Confusion Matrix: Metrics

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

Accuracy

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$\frac{TP + TN}{TP + TN + FN + FP} \tag{15}$$

TNR

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$P[\hat{y} = 0 | y = 0] = \frac{TN}{TN + FP} \tag{16}$$

TPR

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$P[\hat{y} = 1 | y = 1] = \frac{TP}{TP + FN} \tag{17}$$

Example: Unemployment

photo from https://www.dailydot.com/parsec/batman-1966-labels-tumblr-twitter-vine/

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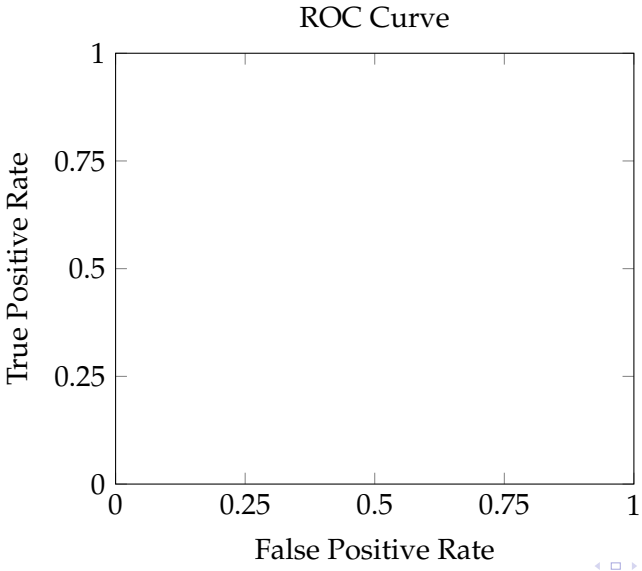
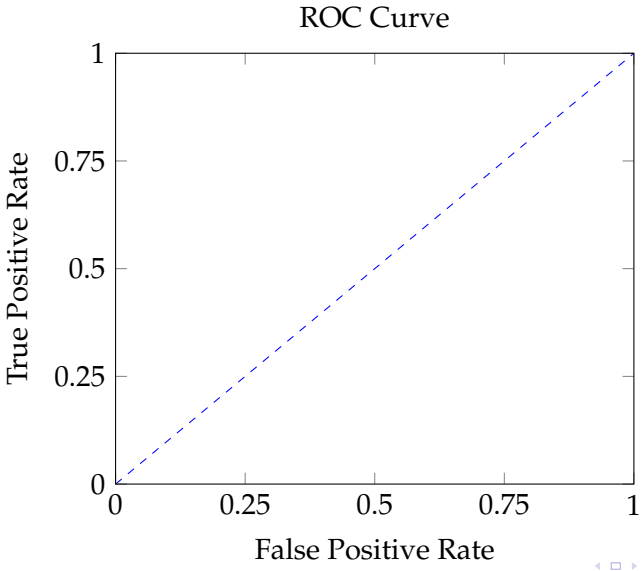
Trade-Off between Different Classification Thresholds

$$\hat{y}_i = 1[p_i \geq c]$$

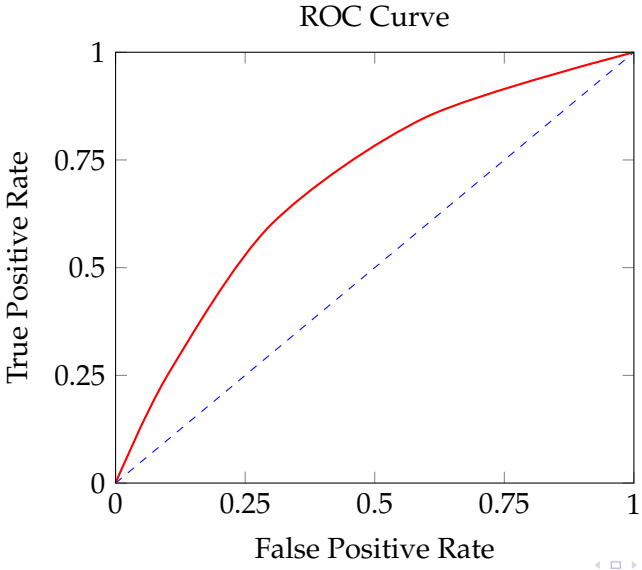
ROC Plot


ROC Curve
1
0.75
0.5
0.25
0
True Positive Rate
0
0.25
0.5
0.75
1
False Positive Rate

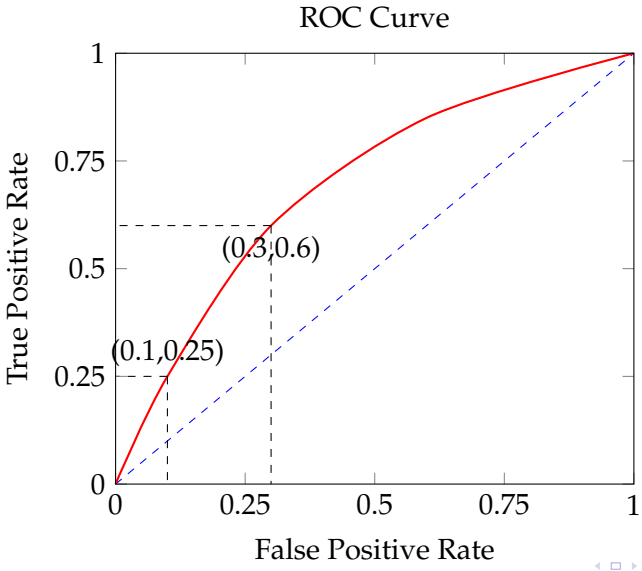
ROC Plot


ROC Curve
True Positive Rate
False Positive Rate
0
0.25
0.5
0.75
1
0
0.25
0.5
0.75
1

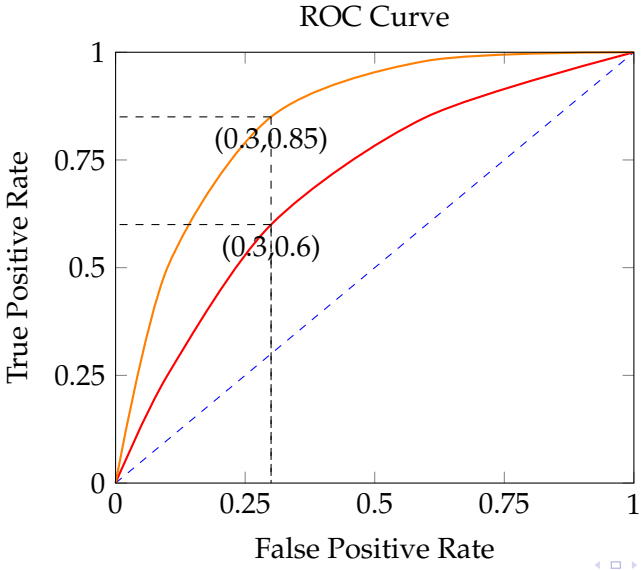
ROC Plot


ROC Curve
True Positive Rate
False Positive Rate
0
0.25
0.5
0.75
1
0
0.25
0.5
0.75
1

ROC Plot


ROC Curve
True Positive Rate
False Positive Rate
0
0.25
0.5
0.75
1
(0.3,0.6)
(0.1,0.25)

ROC Plot


ROC Curve
True Positive Rate
False Positive Rate
0
0.25
0.5
0.75
1
(0.3,0.85)
(0.3,0.6)

Example: Unemployment

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Imbalanced Classification: Motivation

- Interest in one of the classes: Poor, Default, Unemployed, Fraud
- Imbalanced classes pose a challenge

Degree of imbalance	Proportion of Minority Class
Mild	20-40% of the data set
Moderate	1-20% of the data set
Extreme	<1% of the data set

TPR & PPV

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$P[\hat{y} = 1 | y = 1] = \frac{TP}{TP + FN} \tag{18}$$

TPR & PPV

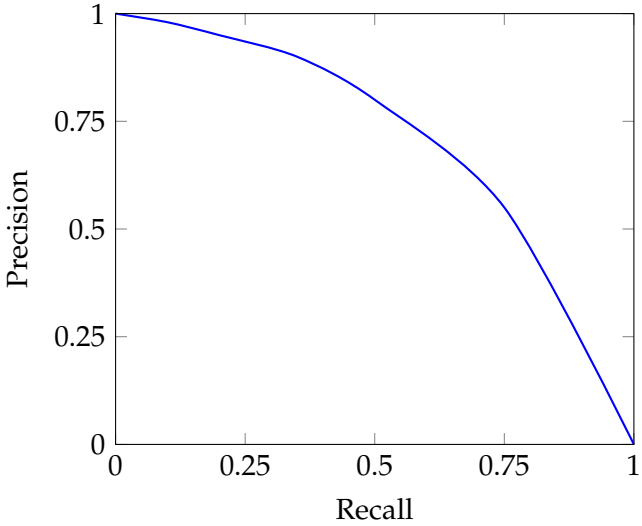
		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$P[\hat{y} = 1 | y = 1] = \frac{TP}{TP + FN} \tag{18}$$

$$P[y = 1 | \hat{y} = 1] = \frac{TP}{TP + FP} \tag{19}$$

PR-Curve

PR-Curve



F-Scores

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall} \tag{20}$$

F-Scores

		y_i	
		1	0
$\hat{y}_i$	1	TP	FP
	0	FN	TN

$$F_\beta = (1+\beta^2)\frac{Precision \times Recall}{(\beta^2 \times Precision + Recall)} \tag{21}$$

Example: Unemployment

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